



Talent Intelligence System

AI-Powered Candidate Ranking, Explained

From job description to ranked shortlist in minutes — built for recruiting teams who need speed without sacrificing trust.

OVERVIEW

From Job Description to Ranked Shortlist in Minutes

The **Talent Intelligence System** transforms how recruiting teams process high-volume candidate pools. Instead of relying on keyword filters or gut-feel shortlisting, the pipeline extracts structured intelligence from every job description and maps it against enriched candidate profiles using a multi-signal scoring engine.

The result: a defensible, ranked shortlist — complete with signal-level transparency and recruiter-ready briefs — delivered in minutes, not days.

JobDNA

Structured extraction from every JD

Multi-Signal

Four weighted scoring layers

LLM Review

Human-like judgment on top-N

Transparent

Every score is explainable

The Problem

Hiring Breaks at Scale

“

Hiring does not scale when candidate volume grows. What works for 200 applicants collapses at 20,000.

”

Keyword Search Over-Ranks Noise

BM25-style matching surfaces keyword stuffers and irrelevant profiles that happen to share terminology, burying genuinely qualified candidates beneath false positives.

Embeddings Miss Hard Constraints

Semantic similarity captures broad relevance but ignores dealbreakers — years of experience, certifications, location — leading to shortlists that fail on must-haves.

Manual Review Collapses

Recruiters spending hours skimming unranked resumes cannot maintain quality at scale. The result: top talent slips through, and review fatigue sets in rapidly.

Recruiters need **ranked recommendations with reasons they can trust** — not another black-box tool that quietly filters out the wrong people.

What I Built – A Six-Layer Pipeline

Each layer addresses a specific failure mode in traditional candidate screening. Together, they form a pipeline that is greater than the sum of its parts.



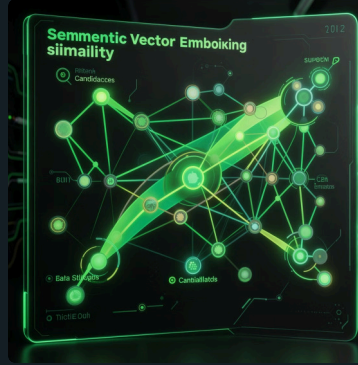
JobDNA Extraction

Parses the job description into a structured object — role, seniority, must-have skills, preferred skills, domain context, and dealbreakers — creating a machine-readable blueprint of what "great" looks like.



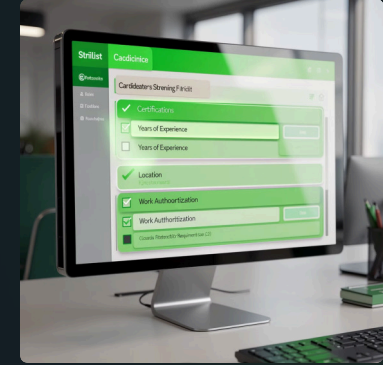
Candidate Profile Enrichment

Transforms raw resume data into recruiter-friendly signals: normalized skills, inferred seniority, career trajectory, and red-flag indicators — making every profile comparable on a level playing field.



Semantic Similarity

Uses embedding-based matching to capture broad relevance — surfacing candidates whose experience aligns conceptually even when exact keywords differ.



Hard-Constraint Matching

Applies strict Boolean-style filtering on must-have requirements — certifications, years of experience, location, work authorization — ensuring no shortlisted candidate fails on dealbreakers.



LLM Judgment

For top candidates only, an LLM applies human-like reasoning — evaluating fit nuances that signals alone cannot capture. Run only on the shortlisted set to control cost and latency.



Trajectory & Red Flags

Rewards upward career momentum while penalizing weak signals: frequent role changes, employment gaps, declining seniority, and missing critical context. A quality-control layer that adds depth to the score.

Design Philosophy

Why This Design – No Single Signal Trusted Alone

The core insight: **no single ranking signal is sufficient on its own**. Each layer compensates for another's blind spots.

Semantic similarity finds relevant profiles but misses hard constraints. BM25 enforces dealbreakers but cannot assess fit nuance. The LLM adds judgment but is too expensive for full-pool scoring. The trajectory layer adds qualitative depth that quantitative signals cannot.

Layered together, the pipeline remains **explainable, affordable, and scalable** – without sacrificing the quality of the final shortlist.



Explainable

Every score decompose into transparent, auditable signal components



Affordable

LLM runs only on top-N, not the full pool



Scalable

Handles 100K+ candidates with deterministic fallbacks

How It Works – End to End

From raw job description to ranked, briefed candidate shortlist — six sequential stages, each building on the previous layer's output.



Ingest Data

Extract JobDNA

Enrich Profiles

Score & Review

The pipeline is designed for modularity — each stage can be swapped, tuned, or bypassed without breaking downstream stages. This makes iteration fast and debugging surgical.

Deliverables

Outputs & UI – Transparency by Design

Every artifact is structured for human review. Recruiters can trace any ranking decision back to its individual signal contributions, and engineering teams can inspect the full analysis for debugging or audit purposes.

1

ranked_candidates.csv

The final shortlist — candidate IDs sorted by composite score, ready for recruiter review. This is the primary output.

2

score_breakdown.csv

Signal-level transparency: each candidate's individual scores for semantic similarity, hard constraints, LLM judgment, and trajectory — so every ranking is fully auditable.

3

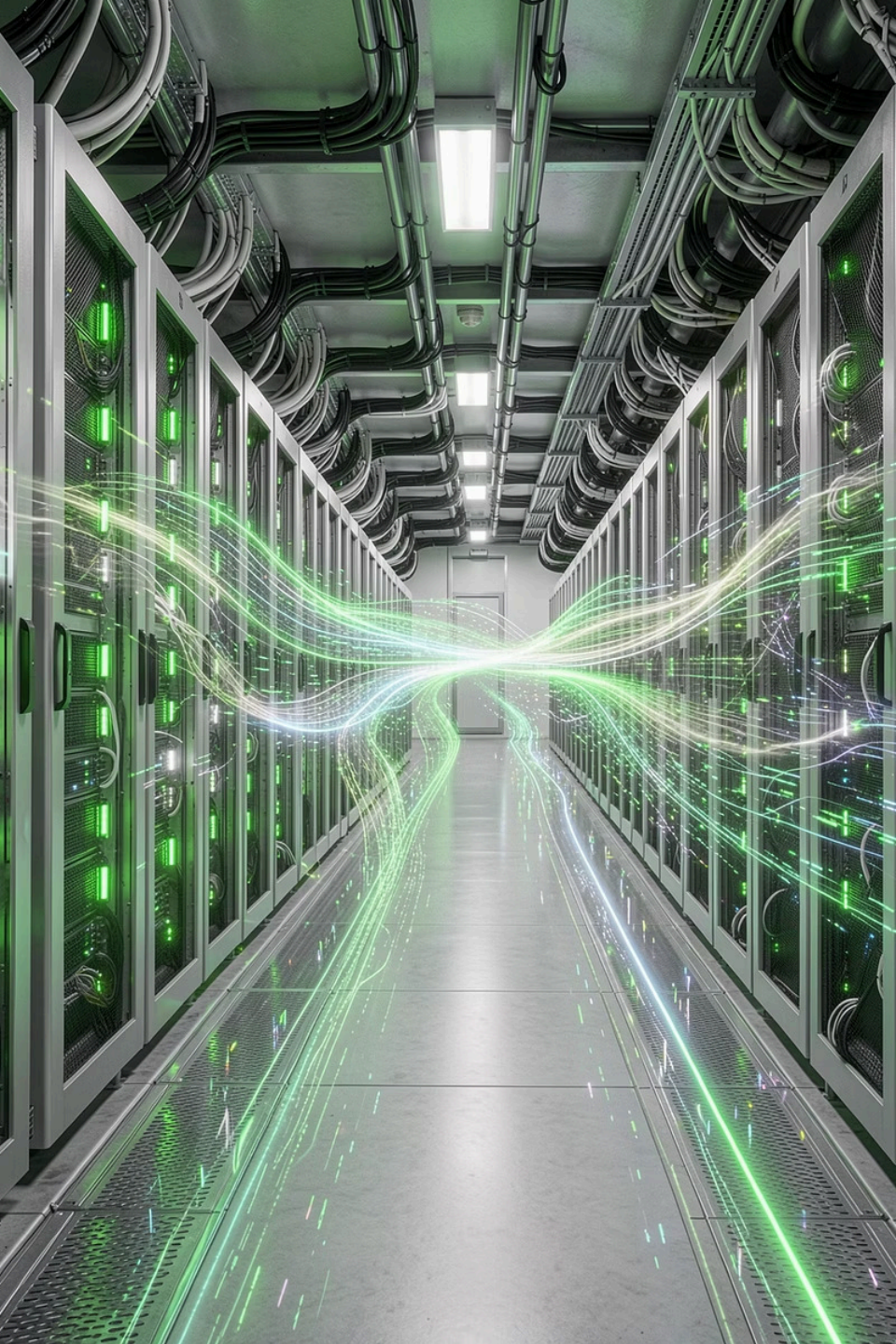
candidate_briefs.csv

Recruiter-ready summaries with key strengths, concerns, and fit rationale per candidate — designed for fast human review without opening every resume.

4

full_analysis.json

Complete structured output for deep inspection, debugging, and integration. Includes every intermediate computation and metadata object.



Example Impact

Validated at 100,000 Candidates

The pipeline has been tested end to end against a dataset of **100,000 candidates**, ranking the full pool with deterministic fallbacks in dry-run mode and with Anthropic API scoring in live mode.

10...

Candidates Ranked

Full pool processed end to
end

4

Weighted Signals

Semantic, constraints,
LLM, trajectory

2

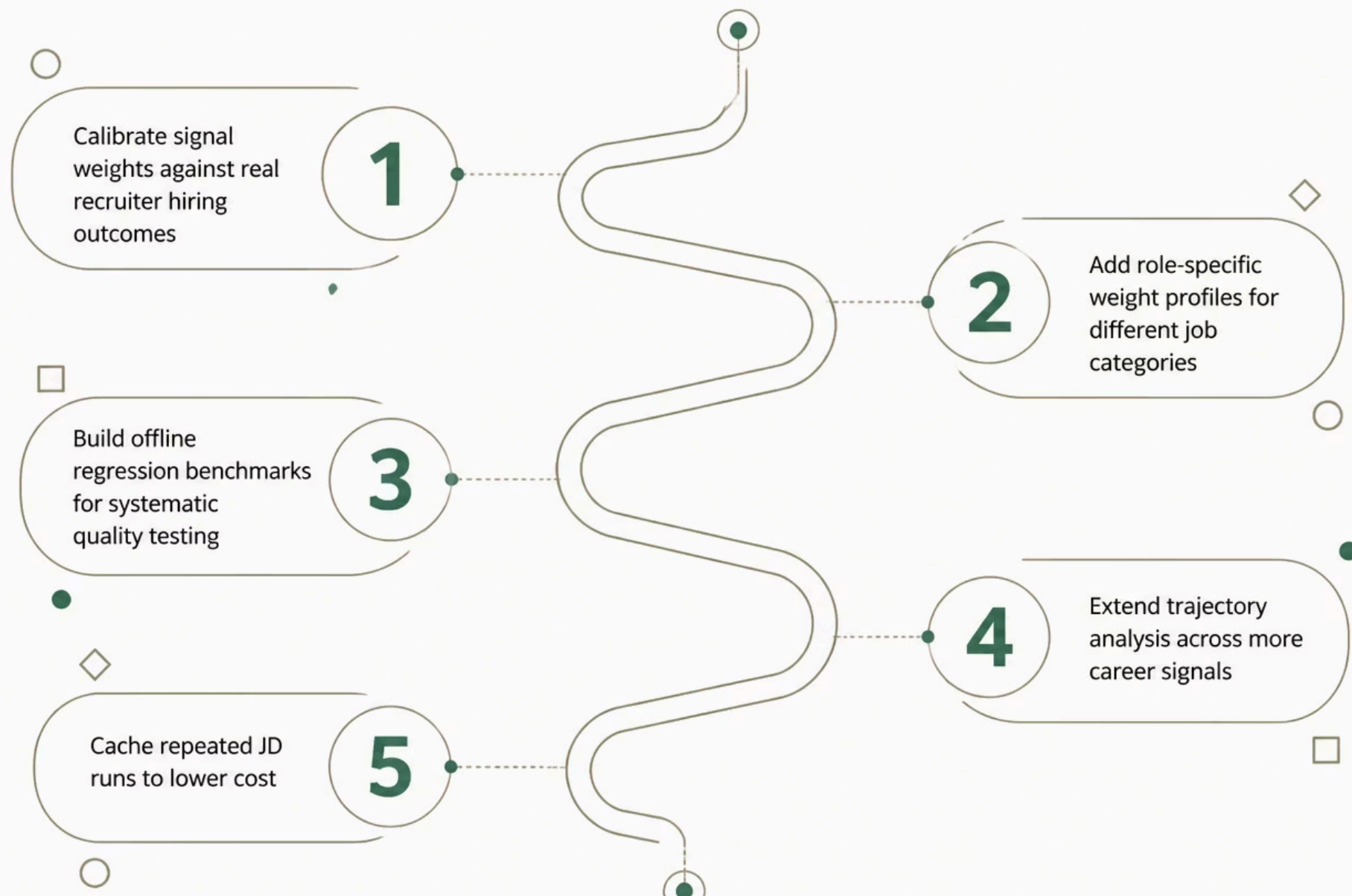
Operating Modes

Dry-run with fallbacks +
live with API

This dual-mode architecture makes the system useful across contexts — from local testing and demos to production-style evaluation against real recruiter outcomes.

What's Next – Calibration and Scale

The current system is a strong foundation. The roadmap focuses on closing the feedback loop, personalizing scoring, and driving down cost per run.



Sample Output

Top Candidates – Ranked Shortlist

Sample results from a dry-run ranking of the candidate pool. Scores reflect the composite of all four weighted signals. Candidates within a 0.01–0.02 range are statistically tied — recruiter judgment drives the final selection.

Rank	Candidate ID	Composite Score
1	CAND_0055905	0.6159 — Dry run mode
2	CAND_0005260	0.6053 — Dry run mode
3	CAND_0072660	0.5947 — Dry run mode
4	CAND_0005649	0.5944
5	CAND_0070525	0.5937 — Dry run mode
6	CAND_0045250	0.5935
7	CAND_0065501	0.5933
8	CAND_0011687	0.5924 — Dry run mode
9	CAND_0094759	0.5919
10	CAND_0081846	0.5916 — Dry run mode

❏ Scores in dry-run mode use deterministic fallbacks for the LLM signal. Live mode with Anthropic API scoring may shift rankings within the tied band — the top tier remains stable.